

Computational Problem Solutions

Chapter 7

$$(2) T \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 2v \\ 0 \\ 5w \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

So $T(v) = Av$ where $A = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$, for $v \in \mathbb{F}^3$

A is upper-diagonal so the eigenvalues are $\lambda = \{0, 0, 5\}$

$\lambda = 0$: For $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$,

$$(A - \lambda I)v = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x \in \mathbb{F} \\ y = 0 \\ z = 0 \end{matrix} \Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$\lambda = 5$:

$$(A - 5I)v = \begin{bmatrix} -5 & 2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -5x + y = 0 \\ 5y = 0 \end{matrix} \Rightarrow \begin{matrix} x = 0 \\ y = 0 \\ z \in \mathbb{F} \end{matrix} \Rightarrow v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Note there are only two eigenvectors. Matrix A is degenerate.

(4)(a) lower triangular $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$, eigs are $\lambda = 3, -1$

$$A - \lambda I = \begin{bmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{bmatrix}$$

$\lambda_1 = 3$: $(A - 3I)v = \begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 8x - 4y = 0 \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = c \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\lambda_2 = -1$: $\begin{bmatrix} 4 & 0 \\ 8 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x = 0 \\ y = y \end{matrix} \Rightarrow v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $c \in \mathbb{R}$

(b) $A = \begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix} \Rightarrow A - \lambda I = \begin{bmatrix} 10 - \lambda & -9 \\ 4 & -2 - \lambda \end{bmatrix}$

$$-(10 - \lambda)(-2 - \lambda) + 36 = (\lambda - 4)^2 = 0 \Rightarrow \lambda = 4, 4$$

$\lambda = 4$: $(A - 4I)v = \begin{bmatrix} 6 & -9 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2x - 3y = 0$

$$\begin{bmatrix} x \\ y \end{bmatrix} = c \begin{bmatrix} 3 \\ 2 \end{bmatrix}, c \in \mathbb{R}$$

eigenvector $v = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \Rightarrow A - \lambda I = \begin{bmatrix} -\lambda & 3 \\ 4 & -\lambda \end{bmatrix}$

$$\lambda^2 - 12 = 0 \Rightarrow \lambda = \pm 2\sqrt{3}$$

$\lambda_+ = 2\sqrt{3}$: $(A - \lambda_+ I)v = \begin{bmatrix} -2\sqrt{3} & 3 \\ 4 & -2\sqrt{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2\sqrt{3}x = 3y \Rightarrow v_+ = \begin{bmatrix} 3 \\ 2\sqrt{3} \end{bmatrix}$

Note: we know these rows are multiples of each other because we selected λ to ensure that!

$\lambda_- = -2\sqrt{3}$: $2\sqrt{3}x + 3y = 0 \Rightarrow v_- = \begin{bmatrix} -3 \\ 2\sqrt{3} \end{bmatrix}$

④ continued

$$(d) A = \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} -(2+\lambda) & -7 \\ 1 & 2-\lambda \end{bmatrix}$$

$$-(2+\lambda)(2-\lambda) + 7 = \lambda^2 + 3 = 0 \Rightarrow \lambda = \pm i\sqrt{3}$$

$$\lambda_+ = i\sqrt{3} \Rightarrow -(2+i\sqrt{3})x - 7y = 0 \Rightarrow V_+ = \begin{bmatrix} 7 \\ 2+i\sqrt{3} \end{bmatrix}$$

$$\lambda_- = -i\sqrt{3}, \quad V_- = \begin{bmatrix} -7 \\ 2+i\sqrt{3} \end{bmatrix}$$

(e) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ clearly has only zero eigenvalue, ($\lambda=0$)

and any $v \in \mathbb{R}^2$ is an eigenvector,

the "0" map (or we can say eigenvectors are $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$)

$$Tx = 0 \in \mathbb{R}^2$$

$$\text{null } T = \mathbb{R}^2$$

(f) $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (the identity map)

$$IX = 1 \cdot X \text{ for any } x \in \mathbb{R}^2$$

so 1 is the eigenvalue - associated with eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

⑤ All of these matrices are upper triangular

(a) eigs are -1, 5

$$A = \begin{bmatrix} -1 & 6 \\ 0 & 5 \end{bmatrix}$$

$$\lambda_1 = -1, v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \leftarrow \text{no brainer}$$

$$A - \lambda I = \begin{bmatrix} -1-\lambda & 6 \\ 0 & 5-\lambda \end{bmatrix}$$

$$\lambda_2 = 5: -6x + 6y = 0 \Rightarrow v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} -1/3 & 0 & 0 & 0 \\ 0 & -1/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix}$$

$$\text{eigs: } (-\frac{1}{3}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}), (-\frac{1}{3}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix})$$

$$(1, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}), (\frac{1}{2}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix})$$

$$(c) A = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & 1/2 & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 1-\lambda & 3 & 7 & 11 \\ 0 & 1/2-\lambda & 3 & 8 \\ 0 & 0 & -\lambda & 4 \\ 0 & 0 & 0 & 2-\lambda \end{bmatrix}$$

$$\lambda_1 = 1, v_1 = (1, 0, 0, 0)$$

$$\lambda_2 = 1/2: A - (\frac{1}{2}I)v = \begin{bmatrix} 1/2 & 3 & 7 & 11 \\ 0 & 0 & 3 & 8 \\ 0 & 0 & -1/2 & 4 \\ 0 & 0 & 0 & 3/2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} w &= 0 \\ z &= 0 \\ y &\in \mathbb{R} \\ \frac{1}{2}x + 3y &= 0 \end{aligned} \Rightarrow \begin{aligned} x &= -3/2 y \\ y &= y \\ z &= 0 \\ w &= 0 \end{aligned} \Rightarrow v = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -3/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} t, t \in \mathbb{R}$$

$$\text{Thus, } \lambda_2 = 1/2, v_2 = (-3, 2, 0, 0)$$

⑤(c) continued

$$\lambda_3 = 0 : Av = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & 1/2 & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} w &= 0 \\ \frac{1}{2}y + 3z &= 0 \\ x + 3y + 7z &= 0 \end{aligned}$$

$$\Rightarrow y = -6z \quad x = 11z$$

$$x = -3y - 7z \Rightarrow y = -6z \Rightarrow v = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 11 \\ -6 \\ 1 \\ 0 \end{bmatrix} z, \quad z \in \mathbb{R}$$

$$= 18z - 7z \quad z = z$$

$$= 11z \quad w = 0$$

$$\boxed{\lambda_3 = 0, v_3 = (11, -6, 1, 0)}$$

$$\lambda_4 = 2: (A - 2I)v = \begin{bmatrix} -1 & 3 & 7 & 11 \\ 0 & -3/2 & 3 & 8 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2z + 4w = 0 \Rightarrow z = 2w$$

$$-3/2 y + 3z + 8w = -3/2 y + 6w + 8w = 0 \Rightarrow y = \frac{28}{3}w$$

$$-x + 3y + 7z + 11w = -x + 28w + 14w + 11w = 0$$

$$\Rightarrow x = -53w$$

$$\text{Then } v = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -53 \\ 28 \\ 2 \\ 1 \end{bmatrix} w, \quad w \in \mathbb{R}$$

$$\boxed{\lambda_4 = 2, v_4 = (-53, 28, 2, 1)}$$

$$\textcircled{6} (a) A = \begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 4-\lambda & -1 \\ 2 & 1-\lambda \end{bmatrix}$$

$$(4-\lambda)(1-\lambda) + 2 = (\lambda-3)(\lambda-2) = 0 \Rightarrow \lambda = 2, 3$$

$$\lambda_1 = 2: 2x - y = 0 \Rightarrow \boxed{v_1 = (1, 2), \lambda_1 = 2}$$

$$\lambda_2 = 3: -1x - 1y = 0 \Rightarrow \boxed{v_2 = (1, -1), \lambda_2 = 3}$$

$$(b) A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ -1 & -\lambda \end{bmatrix} \quad \begin{aligned} \lambda^2 + 1 &= 0 \\ \lambda &= \pm i \end{aligned}$$

$$\lambda_+ = i: -ix + y = 0 \Rightarrow v_+ = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\lambda_- = -i: ix + y = 0 \Rightarrow v_- = \begin{bmatrix} -1 \\ i \end{bmatrix}$$

Note: for real matrix $A \in \mathbb{R}^{n \times n}$, complex eigenvalues/vectors come in conjugate pairs:

$$Av = \lambda v \Leftrightarrow A\bar{v} = \bar{\lambda} \bar{v}$$

$$(c) A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \quad \begin{aligned} &\text{upper triangular} \\ &\Rightarrow \lambda = 2, 2 \end{aligned} \quad A - 2I = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \boxed{v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}} \quad \text{only one eigenvector}$$

$$(d) A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{eigs } \lambda = 1, 0$$

$$\lambda_1 = 1, v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 0, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Math 67 – HW 6 Proof Solutions

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Chapter 7

P8. The claim is **False**.

Consider $T \in \mathcal{L}(\mathbb{R}^2)$ given by $T(x, y) = (y, x)$. This map is clearly invertible, because it is its own inverse. $T = T^{-1}$. The matrix of T with respect to the canonical basis is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ which has all zero diagonal elements.

P9. The claim is **False**.

As an example in $\mathbb{R}^2$, take $T(x, y) = (2x + 6y, x + 3y)$. This operator is not injective since $T(3, -1) = (0, 0)$. Thus, T is not invertible; however, its matrix with respect to the canonical basis for $\mathbb{R}^2$ is $\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$, which has no zeros on the diagonal.

P10. Let V be a finite dimensional vector space over $\mathbb{F}$ and let $S, T \in \mathcal{L}(V)$ be linear operators. Suppose T has $\dim(V)$ distinct eigenvalues and that for every eigenvector $v \in V$ of T with eigenvalue $\lambda \in \mathbb{F}$, the vector v is also an eigenvector of S with an eigenvalue μ . Then $T \circ S = S \circ T$.

Proof. Let $n = \dim(V)$, and let $v_1, v_2, \dots, v_n \in V$ be the eigenvectors of T associated with the distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$, so that $Tv_j = \lambda_j v_j$. Then there exist eigenvalues $\mu_j \in \mathbb{F}$ of S such that $Sv_j = \mu_j v_j$, for $j = 1, 2, \dots, n$. Then for each j ,

$$STv_j = \lambda_j \mu_j v_j = TSv_j.$$

Since the eigenvalues λ_j are distinct, the set $\{v_j\}$ of eigenvectors is linearly independent (by Theorem 7.2.3), so $\{v_j\}$ is a basis for V . Then for any $x \in V$,

$$STx = ST\left(\sum_j \alpha_j v_j\right) = \sum_j \alpha_j STv_j = \sum_j \alpha_j TSv_j = TS\left(\sum_j \alpha_j v_j\right) = TSx,$$

where $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{F}$, and thus $ST = TS$ on V . $\square$

P11. Let V be a finite-dimensional vector space over $\mathbb{F}$ and suppose that $P \in \mathcal{L}(V)$ has the property that $P^2 = P$. Then $V = \text{null}(P) \oplus \text{range}(P)$.

Proof. First we show that $\text{null}(P) \cap \text{range}(P) = \{0\}$: Let $y \in \text{null}(P) \cap \text{range}(P)$. Then $y = Px$ for some $x \in V$, and $P y = 0$. Then $P y = P^2 x = P x = 0$, and since $y = P x$, it follows that $y = 0$.

Next we show that $V = \text{null}(P) + \text{range}(P)$: For any $v \in V$, we have $v = (v - P v) + P v$, where $P v \in \text{range}(P)$ and $v - P v \in \text{null}(P)$, because $P(v - P v) = P v - P^2 v = P v - P v = 0$.

Thus, $V \subseteq \text{null}(P) + \text{range}(P)$, and the opposite inclusion $\text{null}(P) + \text{range}(P) \subseteq V$ follows from the closure of V under addition. $\square$

P12.

(a) For $a, b, c, d \in \mathbb{F}$, the system of equations

$$\begin{aligned} ax_1 + bx_2 &= 0 \\ cx_1 + dx_2 &= 0 \end{aligned} \tag{*}$$

has a non-trivial solution if and only if $ad - bc = 0$.

Proof. ($\Rightarrow$) Assume $(*)$ has a non-trivial solution, i.e. $x_1 \neq 0$ or $x_2 \neq 0$. No generality is lost by only assuming $x_2 \neq 0$, so we only need to consider that case. The system $(*)$ is equivalent to

$$\begin{aligned} acx_1 + bcx_2 &= 0 \\ acx_1 + adx_2 &= 0, \end{aligned}$$

from which we obtain $(bc - ad)x_2 = 0$. Since $x_2 \neq 0$, it follows that $bc - ad = 0$.

($\Leftarrow$) Assume $ad - bc = 0$. Then $(x_1, x_2) = (b, -a)$ is a solution to $(*)$, as is $(-d, c)$. $\square$

(b) Define linear operator $T \in \mathcal{L}(\mathbb{F}^2)$ as $T(v) = Av$, where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{F}^{2 \times 2}$. The eigenvalues of T are $\lambda \in \mathbb{F}$ for which $(a - \lambda)(d - \lambda) - bc = 0$.

Proof. An eigenvalue of A is $\lambda \in \mathbb{F}$ for which $(A - \lambda I)v = 0$ has a non-trivial solution. Equivalently,

$$\begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{aligned} (a - \lambda)v_1 + bv_2 &= 0 \\ cv_1 + (d - \lambda)v_2 &= 0 \end{aligned}$$

has a solution $(v_1, v_2) \neq (0, 0)$. From part a) it follows that such a solution exists if and only if $(a - \lambda)(d - \lambda) - bc = 0$. Then the roots of $p(\lambda) = (a - \lambda)(d - \lambda) - bc$ are exactly the eigenvalues of T . $\square$